

CSE350: Digital Electronics & Pulse Techniques

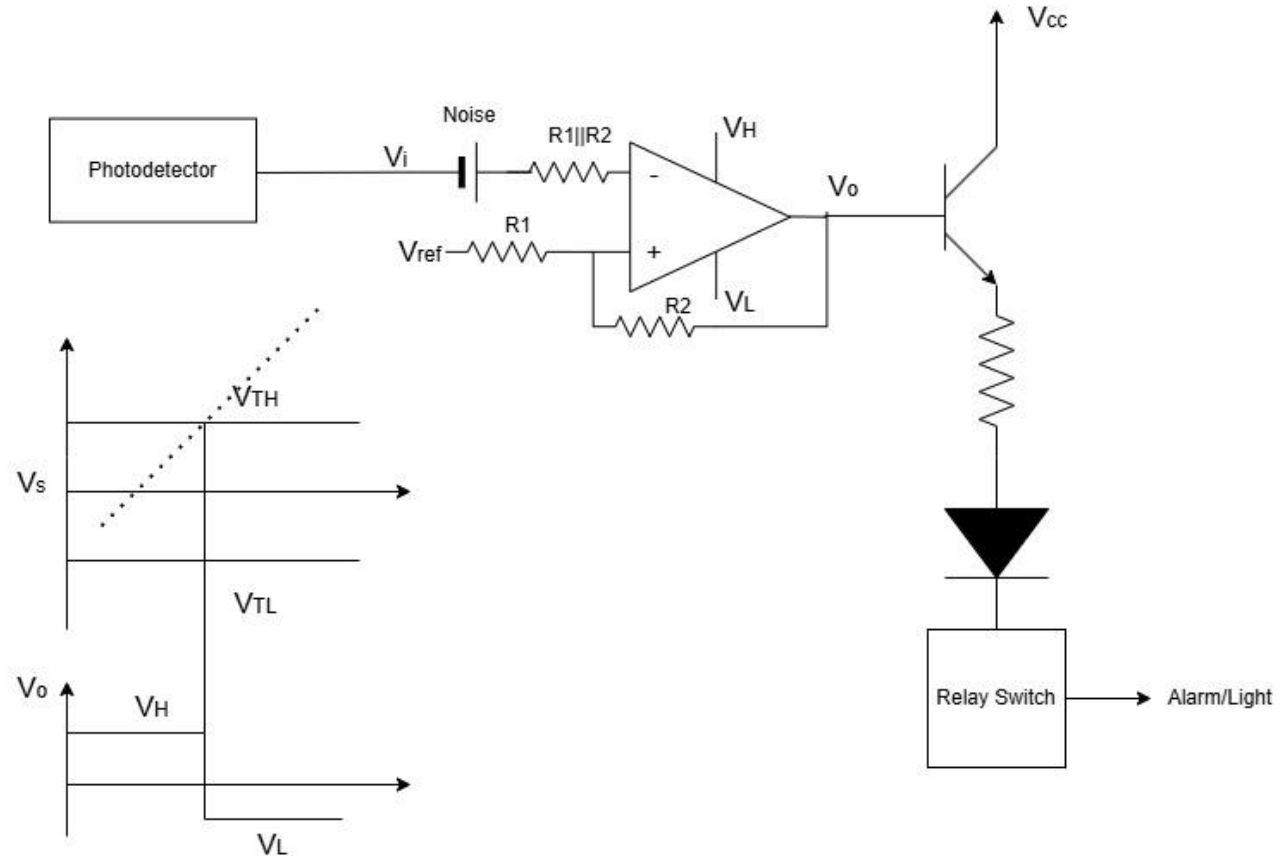
Lecture 12: Applications of Schmitt
Trigger, Signal Generator

Application of Schmitt Trigger:

Street Light/Smoke Detector:

See Design Example 15.7

from Neamen's book.



Schmitt Oscillator/Astable Multivibrator:

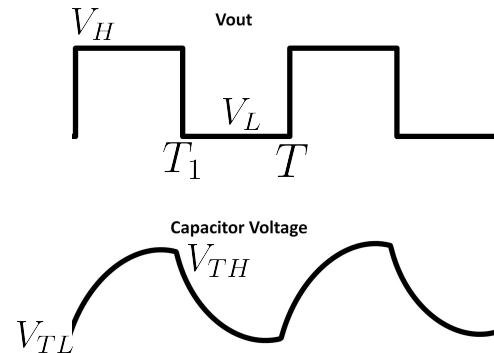
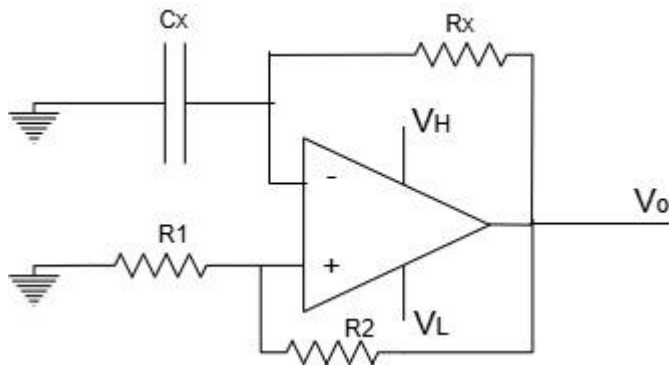
The circuit is effectively an inverting Schmitt Trigger, just that the inverting input is the capacitor voltage. Now, since no current flows through the terminals of the op-amp, R_x & C_x can be considered forming a single series loop with output voltage. This is an RC circuit, and the voltage across the capacitor can be expressed as,

$$V_{cx}(t) = V_o + [V_{cx}(0) - V_o]e^{-\frac{t}{R_x C_x}}$$

Let's assume that, $V_{cx}(0) = V_{TL}$

In such case, the output voltage will be V_H

We want to find the time after which the voltage of the capacitor reaches V_{TH} , T_1



Applying these initial conditions to the capacitor voltage equation, we get,

$$V_{TH} = V_H + [V_{TL} - V_H]e^{-\frac{T_1}{R_x C_x}}$$

$$\Rightarrow e^{-\frac{T_1}{R_x C_x}} = \frac{V_{TH} - V_H}{V_{TL} - V_H}$$

$$\Rightarrow -\frac{T_1}{R_x C_x} = \ln\left(\frac{V_{TH} - V_H}{V_{TL} - V_H}\right)$$

$$\Rightarrow T_1 = R_x C_x \ln\left(\frac{V_{TL} - V_H}{V_{TH} - V_H}\right)$$

Now, we can do the same analysis, assuming initial conditions at T_1 , and final value at T . We get,

$$V_{TL} = V_L + [V_{TH} - V_L]e^{-\frac{T-T_1}{R_x C_x}}$$

$$\Rightarrow e^{-\frac{T-T_1}{R_x C_x}} = \ln\left(\frac{V_{TL} - V_L}{V_{TH} - V_L}\right)$$

$$\Rightarrow T - T_1 = R_x C_x \ln\left(\frac{V_{TH} - V_L}{V_{TL} - V_L}\right) \Rightarrow T = T_1 + R_x C_x \ln\left(\frac{V_{TH} - V_L}{V_{TL} - V_L}\right) \quad \text{Or,} \quad T_2 = R_x C_x \ln\left(\frac{V_{TH} - V_L}{V_{TL} - V_L}\right)$$

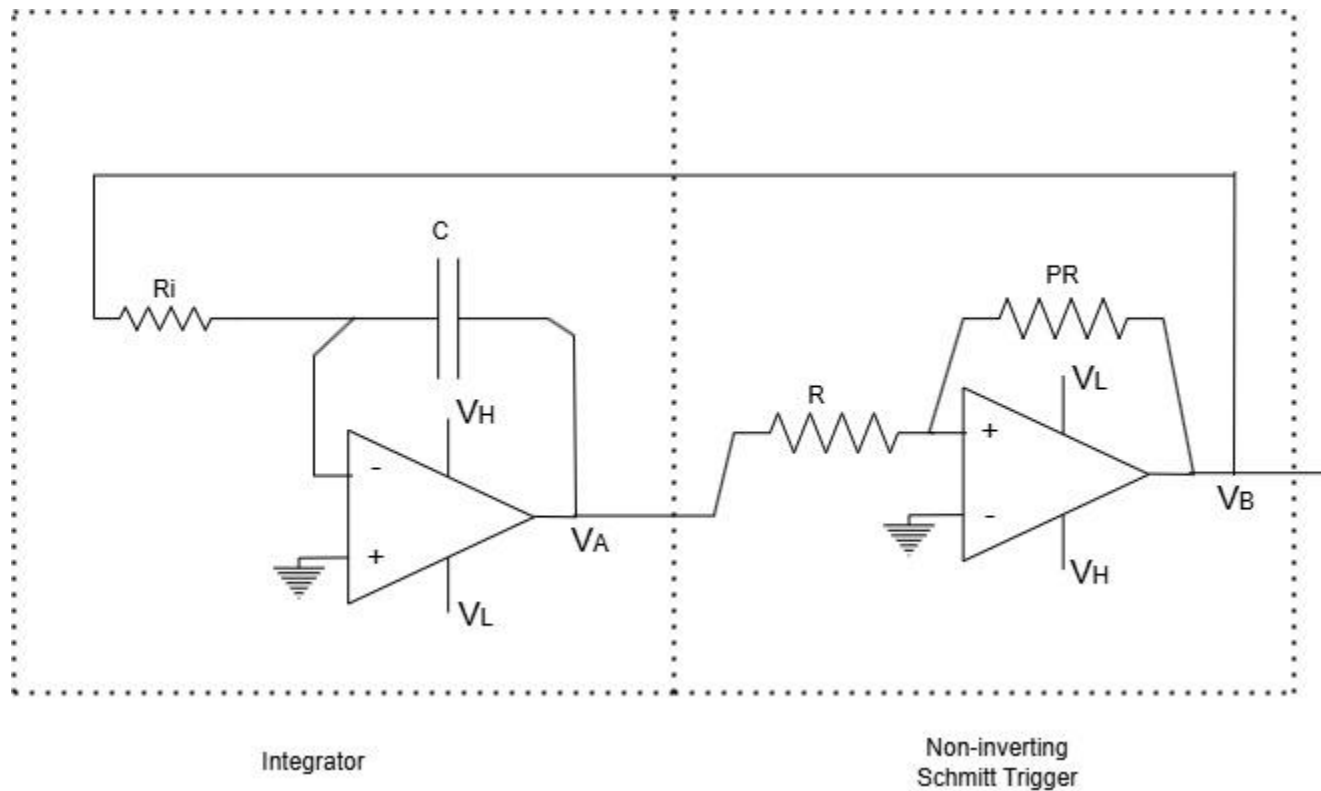
Here T_2 is the time during which the output square wave is at its low peak (V_L).

The **time period** is T . Thus, the **frequency** of the square wave is, $f = \frac{1}{T}$

The **duty cycle** is the percentage of time period, during which the square wave is at its high peak (V_H).

So, duty cycle, $D = \frac{T_1}{T} * 100$

Triangular Wave Generator:



Since, voltage at both terminals of the integrator is 0, the output voltage of the integrated circuit is the voltage across the capacitor. Thus, we can write,

$$V_c(t) = -\frac{V_s}{R_i C} \int_{t_1}^t dt$$

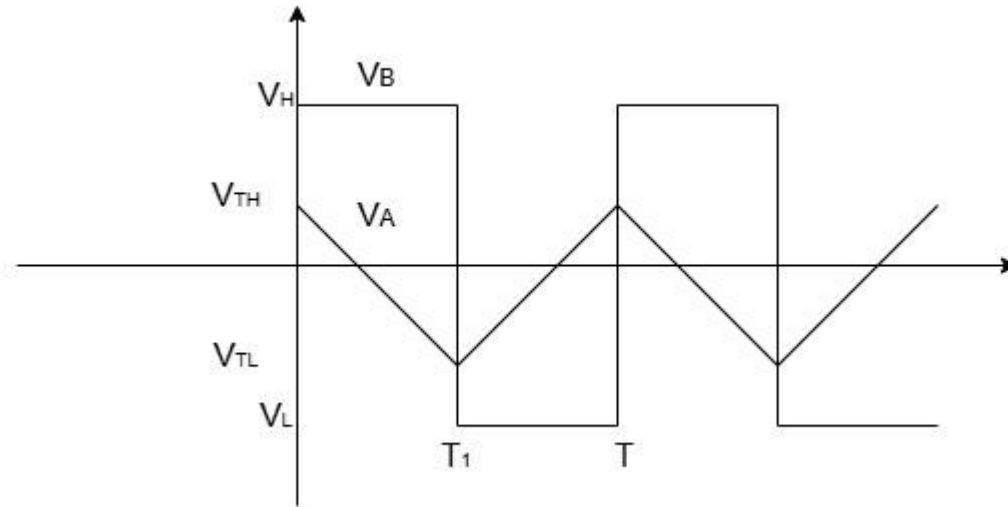
$$V_c(t) = V_c(t_1) - \frac{V_s}{R_i C} (t - t_1)$$

For the non-inverting Schmitt trigger, we can write,

$$V_{TH} = -\frac{V_L R}{PR} = -\frac{V_L}{P}$$

$$V_{TL} = -\frac{V_H R}{PR} = -\frac{V_H}{P}$$

Output of integrator & non-inverting Schmitt trigger:



Determination of T_1 [Case: $0 < t < T_1$]

$$V_c(T_1) = V_c(0) - \frac{V_B}{R_i C} (T_1 - 0) \longrightarrow V_{TL} = V_{TH} - \frac{V_H}{R_i C} T_1 \longrightarrow T_1 = R_i C \frac{V_{TH} - V_{TL}}{V_H}$$

Case: $T_1 \ll T$

Similar analysis gives,

$$T - T_1 = T_2 = R_i C \frac{V_{TL} - V_{TH}}{V_L}$$

$$T = T_1 + T_2$$

$$f = \frac{1}{T}$$

Duty Cycle,

$$D = \frac{T_2}{T_1 + T_2}$$